

Sai Karthik Kagolanu

AI and ML Engineer

Sheffield, UK | 07775 439150 | sai.kagolanu@yahoo.com | Graduate visa to Feb 2027

GitHub | LinkedIn | Portfolio

SUMMARY

MSc Robotics graduate (Sheffield) who builds AI agents, ML pipelines and production backends. Raised billing agent containment from 41% to 74% at Fuse Energy and saved 23% of wasted engineer callouts. Built Trezia, the accounting platform a 55-flat housing society runs its books on. Two hackathon wins.

SKILLS

Languages: Python, C++, SQL, JavaScript/TypeScript

ML and data: PyTorch, TensorFlow, scikit-learn, XGBoost, ONNX Runtime, MLflow, LangChain, pandas, OpenCV

Backend and infrastructure: FastAPI, PostgreSQL, Redis, Kafka, Docker, dbt, Dagster, Prometheus/Grafana, GitHub Actions

Other: React/Next.js, Supabase, Cloudflare R2, BigQuery, ClickHouse, AWS, Git

EXPERIENCE

Operations Intern

Fuse Energy

Mar 2026 – Jun 2026

London, UK

- Raised the billing agent's autonomous containment from **41% to 74%** in production by growing its test suite from 63 to 103 cases targeting tool-call leakage and context drift.
- Cut unnecessary field visits by **23%** with queries and dashboards that flagged metering and install faults early (ClickHouse, Metabase).
- Removed mid-call documentation searches by shipping an internal AI assistant that answered support staff questions directly from internal docs.
- Traced recurring escalations to two root causes and rewrote both workflows; resolved second-line cases and prepared ombudsman case files.

Machine Learning Intern

Infyz Solutions

Jun 2023 – Aug 2023

Hyderabad, India

- Cut model training time by **40%** by rewriting preprocessing across three production classifiers (pandas, NumPy).
- Picked the production model by benchmarking Random Forest, SVM and XGBoost on 5,000+ labelled samples; it deployed at **92% accuracy**.

PROJECTS

Trezia | Live

Jun 2026 – Present

- Replaced a 55-flat society's four-year Excel workbook with a live accounting platform used daily by a non-technical treasurer: billing, dues tracking, PDF receipts, balance sheets and an immutable audit trail (Next.js, Supabase; 424 tests).
- Runs multi-society on one deployment. Made collections free through one-tap UPI and WhatsApp payment links, with optional per-society Razorpay, OCR-checked payment proofs, a resident portal and full CSV/JSON export.
- Let the treasurer query the books in plain English through a read-only AI assistant with four query tools (Grog).

Axiar | Live

Jan 2026 – Present

- Built a multi-tenant, event-driven platform for bookings, dispatch and payments with 358 passing tests and CI/CD (FastAPI, PostgreSQL/PostGIS, Redis, Kafka, React).
- Gave operators plain-language answers on revenue, demand and bookings through a tool-calling agent over live tenant data (LangChain, local LLM).

- Fixed six security-audit findings — tenant data isolation, JWT type confusion, GPS spoofing — with a regression suite now in CI, and enforced booking-overlap rules at the database level.

Industrial Intelligence Platform | Code

Jun 2026

- Built a predictive maintenance platform where simulated equipment telemetry feeds failure classification across 10 Docker services (XGBoost, asyncio).
- Automated daily retraining with model promotion gated on F1 above 0.70 (champion at 0.737 macro-F1), with feature engineering in dbt and full observability (Dagster, MLflow, Prometheus, Grafana).

RAG Demo | Code | Live

Jan 2026 – Feb 2026

- Answered welding-defect questions **30%** more accurately than keyword search, in under four seconds, by fusing keyword and semantic retrieval with cross-encoder reranking.
- Deployed on Streamlit Cloud with query transformation, faithfulness scoring, semantic caching and **96%** test coverage.

Computer Vision Classification Suite | Code

Nov 2025 – Mar 2026

- Classified images at **88.9%** accuracy across six classes on 14,034 photos, with a C++ inference engine running **2–3x** faster than Python (ONNX Runtime).
- Served predictions through a monitored backend and React dashboard at **93%** test coverage (FastAPI, Prometheus, Grafana).

EDUCATION

MSc Robotics, Merit
University of Sheffield

Sep 2023 – Sep 2024
Sheffield, UK

Dissertation: gesture and speech-controlled robotic arm with **96.98%** gesture accuracy and **95.03%** speech accuracy using custom CNN architectures. Modules: Machine Vision, Machine Intelligence, Deep Learning.

BTech Mechanical Engineering, 8.84/10 (equiv. First Class)
Vellore Institute of Technology

Jul 2019 – May 2023
Chennai, India

Dissertation: Surgical Collaborative Robotic Arm.

AWARDS

Best Use of AI Agents on Arm – HackSheffield10 (2025). Five-agent shopping assistant over live Frasers Group stock, containerised on Arm hardware inside 24 hours.

Winner – Sheffield AI Hackathon (May 2025). *SheffAware*: livability scores for Sheffield's 28 wards using open council data. Live at sheffaware.streamlit.app.